

Easyflow by Neles™ K-series limit switches

The Easyflow by Neles K-series features versatile range of limit switches for quarter turn actuators and valve assemblies. With the K-series limit switches our customers receive reliable information of the limits of desired rotary travel set by the customer. Limit switch shaft is coupled directly to the actuator shaft with NAMUR (VDI/VDE 3845) adaptation. Actuator movement is transferred via the unit shaft to preset cams that actuate either inductive or mechanical switch elements. The switches are prewired to terminal strips for easy connection to an external monitoring system.



BENEFITS

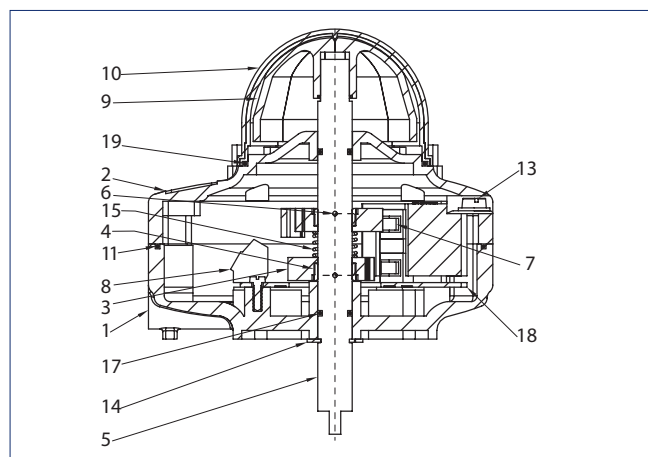
- The versatile K-series limit switches can be mounted on top of virtually any rotary actuator
- - VDI/VDE 3845 mounting face
- K-series limit switch is very cost competitive solution for sensing the limits of rotary travel
- Its compact size makes it easy to fit even to tight spaces
- K-series limit switches are suitable even for the most harsh weather conditions with three different housing material options and IP67 ingress protection class as standard
- Switch elements can be chosen between mechanical, inductive and reed type switches
- SOV termination provided inside the limit switch box (with D, E, F, H, R and V switch elements)

FEATURES

- Two product series
 - KS = K-series, standard housing
 - KC = K-series compact housing
- Three housing material options
 - Epoxy powder coated, low copper die cast aluminum (LM6)
 - Stainless steel (CF8M, equivalent to AISI316)
 - Polycarbonate (LEXAN 3412R) (only for KC-series)
- Cable entry options
 - M20x1,5
 - ½"NPT
 - ¾"NPT (KS only)
- Choice of 9 switch element variants
 - Inductive
 - Mechanical
 - Reed type
- Large visual indicator
 - Inside a shatter proof polycarbonate dome
- VDI/VDE 3845 mounting face with NAMUR shaft
- Four different temperature ranges
 - General: -20 °C to +80 °C
 - Cold: -40 °C to +80 °C
 - Arctic: -50 °C to +80 °C
 - High: -20 °C to +100 °C
- KS-series with two or four cable entries and KC-series with three cable entries
 - M20x1.5, ½"NPT, or ¾"NPT (Only with KS)
- Easily adjustable cams
- Cams color coded to ease up installation
- IP67 as standard
- Switch elements connected to terminals via PCB to ensure good short circuit protection

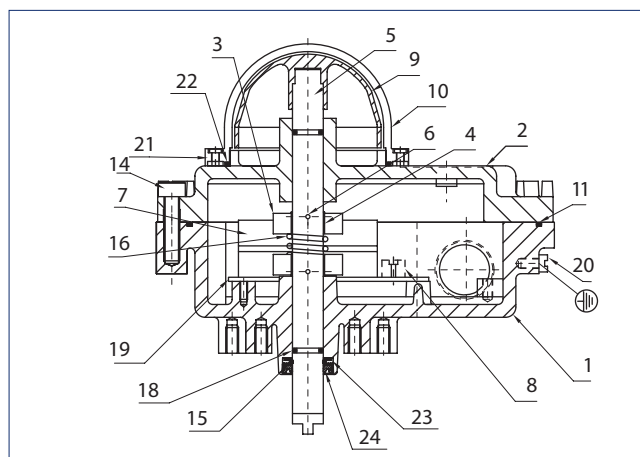
ASSEMBLY DRAWING AND MAIN PARTS

Series KC



S.NO.	DESCRIPTION	QTY.
1	HOUSING	1
2	COVER	1
3	CAM	2
4	CAM-LOCK 2	2
5	SHAFT	1
6	SPRING DOWEL	2
7	SWITCH ELEMENTS	2
8	TERMINAL BLOCK	4
9	INDICATOR	1
10	INDICATOR COVER	1
11	GASKET	1
13	NON FALLING SCREW	4
14	'E' TYPE CIRCLIP	1
15	SPRING	1
17	O-RING	2
18	PRINTED CIRCUIT BOARD	1
19	O-RING	1

Series KS



S.NO.	DESCRIPTION	QTY.
1	HOUSING	1
2	COVER	1
3	CAM	2
4	CAM-LOCK	2
5	SHAFT	1
6	SPRING DOWEL	2
7	SWITCH ELEMENTS	2
8	CONNECTOR STRIP	1
9	INDICATOR	1
10	INDICATOR COVER	1
11	GASKET	1
14	NON FALLING SCREW	8
15	EXTERNAL CIRCLIP	1
16	SPRING	1
18	O-RING	2
19	PRINTED CIRCUIT BOARD	1
20	SCREW FOR EARTH CONN.	2
21	PHILIPS HEAD SCREW	4
22	O-RING	1
23	WASHER	1
24	CUP SEAL	1

OPERATION PRINCIPLE

The K-series limit switches include from two to four inductive, mechanical or reed type switch elements. They are activated by cams that are directly driven by the actuator shaft through the limit switch shaft. Cams are set by the customer to the desired rotary travel limits. The switch elements are prewired to terminal

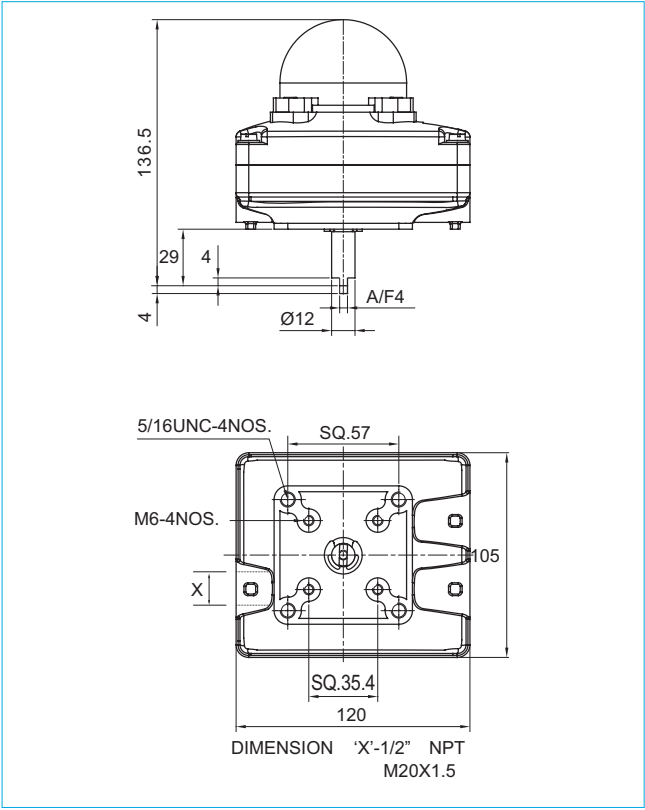
strip in the enclosure, permitting easy connection of switch output to external electrical monitoring system or indication devices. K-series limit switches also include a visual indicator on top of the enclosure cover that indicates 90° rotary travel between open and closed valve position.

TECHNICAL SPECIFICATION

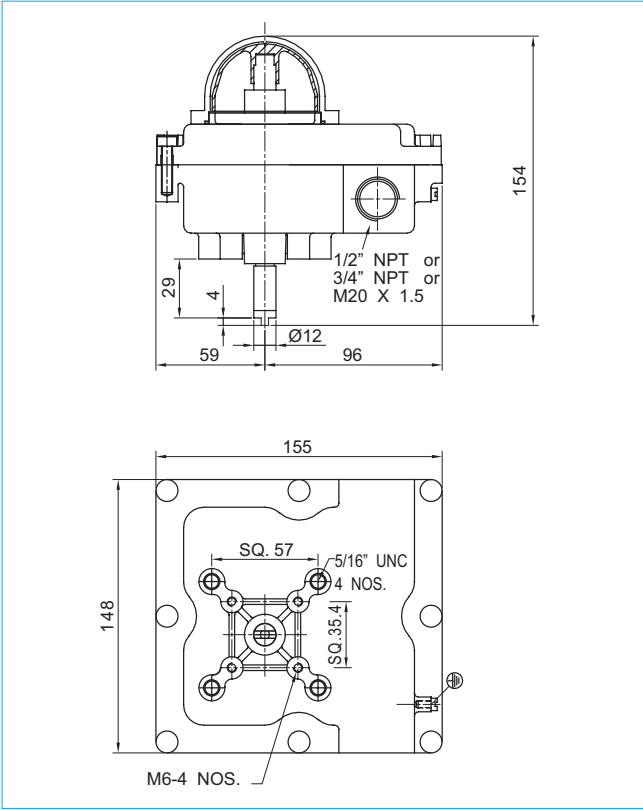
Ambient temperature:		D1 :	Inductive proximity switch, P+F NJ2-V3-N 2-wire type, Namur NC
General:	-20 ... +80 °C / -4 ... +176 °F	Supply voltage:	5 - 25 V DC, nominal 8 V DC (Ri = approx. 1 kOhm)
Cold:	-40 ... +80 °C / -40 ... +176 °F	Output current:	
Arctic:	-50 ... +80 °C / -58 ... +176 °F	Active face free:	> 3 mA
High:	-20 ... +100 °C / -4 ... +212 °F (Not applicable with polycarbonate body)	Active face covered:	< 1 mA
Protection class:	IP67	Nominal sensing range:	2 mm
Body:		D2 :	Inductive proximity switch, P+F NBB3-V3-Z4 2-wire type, NO
KS series:	Epoxy powder coated, low copper die cast aluminum (LM6) / stainless steel (CF8M)	Supply voltage:	5 - 60 V DC
KC series:	Epoxy powder coated, low copper die cast aluminum (LM6) / stainless steel (CF8M) / polycarbonate (LEXAN 3412R)	Output current:	
Pointer cover:	Shatter proof polycarbonate (INFINO SC-1220UR)	Operating current:	4 - 100 mA
Internal parts:		Off-state current:	0,4 - 0,55 mA
Shaft:	AISI304 / AISI316 (SS body)	Nominal sensing range:	3 mm
Cams:	ABS	F1:	Inductive proximity switch, P+F SJ3.5-N 2-wire type, Namur NC
PCB:	Grade FR4	Supply voltage:	5 - 25 V DC, nominal 8 V DC (Ri = approx. 1 kOhm)
Fasteners:	AISI304	Output current:	
Sealing:	Nitrile rubber (NBR) (General temp range) Silicon (Cold temp range) Fluoro silicon (Arctic temp range) Viton (High temp range)	Active face free:	> 3 mA
Conduit entry:		Active face covered:	< 1 mA
KS series:	2 x M20 or 2 x 1/2 NPT or 2 x 3/4 NPT 4 x M20 or 4 x 1/2 NPT or 4 x 3/4 NPT	Slot width:	3,5 mm
KC series:	3 x M20 or 3 x 1/2 NPT	F2 :	Inductive proximity switch, P+F SJ3.5-SN 2-wire type, Namur NC
Weight:		Supply voltage:	5 - 25 V DC, nominal 8 V DC (Ri = approx. 1 kOhm)
KS series:	Die cast aluminum 1.7 kg / 3.8 lbs, stainless steel 4.5 kg / 9.9 lbs	Output current:	
KC series:	Die cast aluminum 0.7 kg / 1.6 lbs, stainless steel 2.5 kg / 5.5 lbs, polycarbonate 0.6 kg / 1.3 lbs	Active face free:	> 3 mA
Switches:		Active face covered:	< 1 mA
A1:	Inductive proximity switch, P+F NJ2-12GM-N 2-wire type, Namur NC	Slot width:	3,5 mm
Supply voltage:	5 - 25 V DC, nominal 8 V DC (Ri = approx. 1 kOhm)	R1:	Reed proximity switch, Metso R1 220 V AC/DC
Output current:		Supply voltage:	
Active face free:	> 3 mA	Output current:	2000 mA
Active face covered:	< 1 mA	Operating current:	
Nominal sensing range:	2 mm	Off-state current:	0 mA
A3 :	Inductive proximity switch, P+F NBN4-12GM40-Z0 2-wire type, NO	V1:	Mechanical micro switch, Honeywell V15S05-CZ100A05-01, SPDT type
Supply voltage:	5 - 60 V DC	Electrical values:	5 A @ 125/250 VAC, 100 mA @ 48 VDC or 125/250 VAC
Output current:		W1:	Mechanical micro switch, Honeywell V15S05-CZ100A05-01, DPDT type
Operating current:	2 - 100 mA	Electrical values:	5 A @ 125/250 VAC, 100 mA @ 48 VDC or 125/250 VAC
Off-state current:	0 - 0,5 mA		
Nominal sensing range:	4 mm		

DIMENSIONAL DRAWINGS, mm

Series KC



Series KS



Easyflow by Neles™ K-series limit switches type code

1. sign									PRODUCT GROUP
KC									Compact enclosure, weather proof
KS									Standard enclosure, Ex d capable
2. sign									QUANTITY OF SWITCHES
2									2 switches
3									3 switches
4									4 switches
3. sign									SWITCH TYPE
									INDUCTIVE PROXIMITY SWITCHES
									Manufacturer Model Note
A1									P&F NJ2-12GM-N Cylindrical
A3									P&F NBN4-12GM40-Z0 Cylindrical
D1									P&F NJ2-V3-N Flat
D2									P&F NBB3-V3-Z4 Flat
F1									P&F SJ3.5-N Slot
F2									P&F SJ3.5-SN Slot
									Electrical values
									8V DC, 3mA, NC
									10-60V DC, 100mA, NO
									8 VDC, >3 mA ; <1 mA (NC)
									8 VDC, >3 mA ; <1 mA (NC)
									REED TYPE PROXIMITY SWITCHES
									Manufacturer Model Note
R1									Metso R1 SPDT, Not applicable with 1. sign KC
									Only applicable with 7. sign G and C
									MECHANICAL MICRO SWITCHES
									Manufacturer Model Note
V1									Honeywell V15S05-CZ100A05-01 SPDT
W1									Honeywell V15S05-CZ100A05-01 DPDT
									Not applicable with 1. sign KC
									Only applicable with 2. sign 2"
4. sign									ENCLOSURE MATERIAL
A									Diecast aluminum (LM6)
S									Stainless steel CF8M (AISI 316 / CF8M)
P									Only with 1. sign KC
									Polycarbonate (LEXAN 3412R)
5. sign									QUANTITY OF CONDUIT ENTRIES
2									Only with 1. sign KS
3									2 entries
4									Only with 1. sign KC
									3 entries
									Only with 1. sign KS
									4 entries
6. sign									CABLE ENTRY
M									Only with 1. sign KS
N									M20x1.5
L									1/2NPT
									3/4NPT
7. sign									TEMPERATURE RANGE
G									Only with 3. signs F2, R1, V2 and W2
C									Only with 3. sign F2
A									Not applicable with 8. sign EN
H									Not applicable with 4. sign P
									Only applicable with 3. signs A1, D1, F1, F2 and R1
									With aluminum indicator, no polycarbonate dome
									Only to temperature class T4
8. sign									VISUAL INDICATOR
R									Red and yellow
G									Red and green
9. sign									APPROVALS
NN									Only with 1. sign KS
EN									No approvals
									ATEX / IECEx Ex d
KS	2	A1	A	2	M	G	R	NN	

Neles

Vanha Porvoontie 229, 01380 Vantaa, Finland.
Tel. +358 10 417 5000.

neles.com

S100-1EN - 7/2020

Subject to change without prior notice. Neles, Jamesbury and Easyflow by Neles, and certain other trademarks, are either registered trademarks or trademarks of Neles Corporation or its subsidiaries or affiliates in the United States and/or in other countries.

Reinventing
reliability